3031ICT: Applied Data Mining

3804ICT: Data Mining

Lecture 10: World-Wide-Web Mining



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World-Wide-Web Mining

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- Web Usage Mining
  - Types of Data
  - Web Usage Data Pre-Processing
  - Web Usage Mining: Modelling
- Recommender Systems and Collaborative Filtering
  - Collaborative Filtering
    - Finding “Similar” Users
    - Rating Predictions
  - Item-Item Collaborative Filtering
  - Item-Item vs. User-User
  - Pros/Cons of Collaborative Filtering

Web Usage Mining: Background

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- A large amount of **clickstream**, **transaction data**, and **user profile data** are collected by Web-based organizations (e-commerce) in their daily operations
- Analyzing such data can help these organizations:
  - **determine** the life-time value of clients,
  - **design** cross-marketing strategies across products and services,
  - **evaluate** the effectiveness of promotional campaigns,
  - **optimize** the functionality of Web-based applications,
  - **provide** more personalized content to visitors,
  - and **find** the most effective logical structure for their Web space.

This type of analysis involves the automatic discovery of **meaningful patterns** and **relationships** from a large collection of primarily semi-structured data, often stored in Web and applications server access **logs**, as well as in related operational data sources.

Web Usage Mining

- **Web usage mining** refers to the automatic discovery and analysis of patterns in clickstreams, user transactions and other associated data collected or generated, as a result of user interactions with Web resources on one or more Web sites.
- **The goal** is to **capture**, **model**, and **analyze** the behavioral patterns and profiles of users interacting with a Web site.
- The **discovered patterns** are usually represented as **collections of pages**, **objects**, or **resources** that are frequently accessed or used by groups of users with common needs or interests.

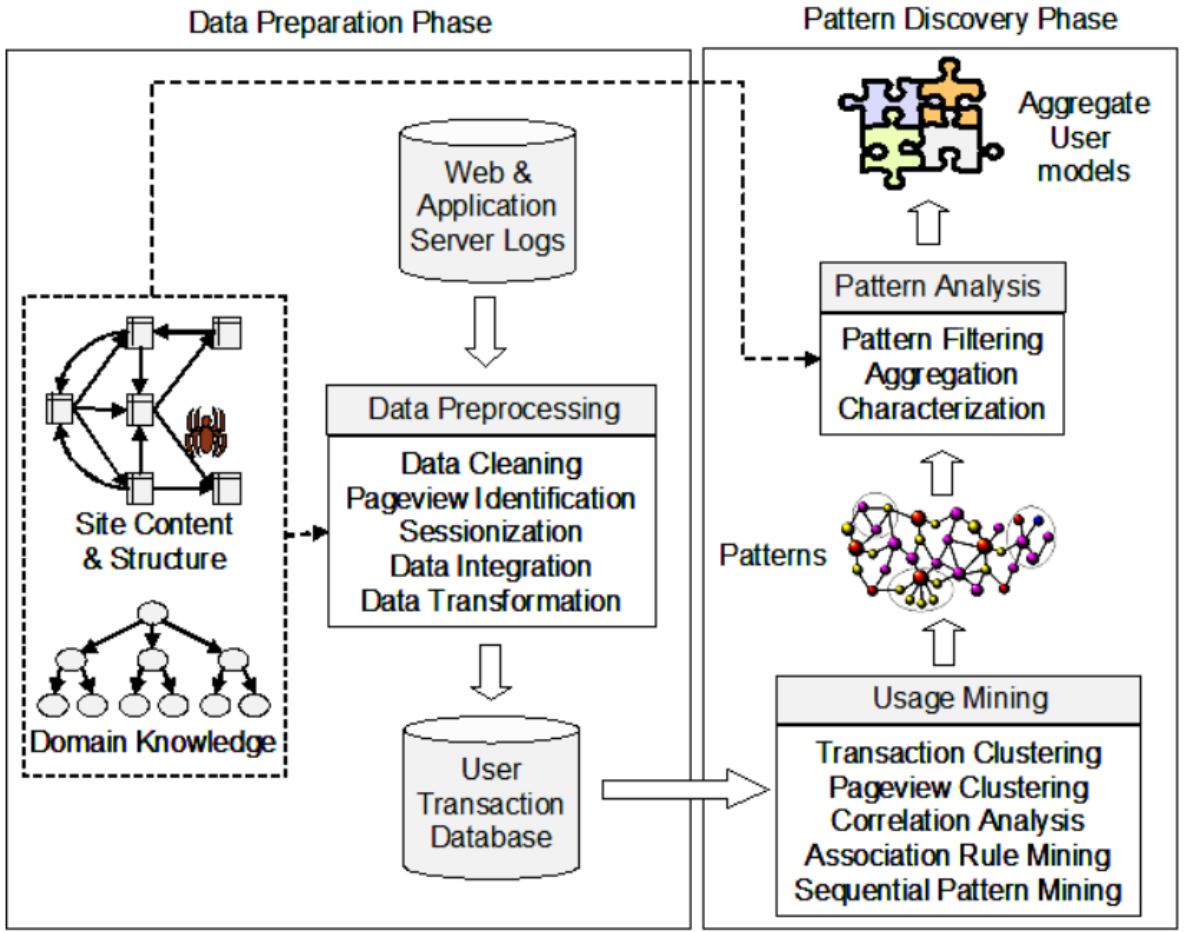
Web Usage Mining

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The overall Web usage mining process can be divided into **three inter-dependent stages**:

- Data collection and pre-processing
  - the **clickstream data** is cleaned and partitioned into a set of user transactions representing the activities of each user during different visits to the site.
  - Other **sources of knowledge** such as the site content or structure, as well as semantic domain knowledge from site ontologies (such as product catalogs or concept hierarchies), may also be used in pre-processing, or to enhance user transaction data.
- Pattern discovery
  - **statistical**, **database**, and **machine learning** operations are performed to obtain **hidden patterns** reflecting the typical behavior of users, as well as summary statistics on Web resources, sessions, and users.
- Pattern analysis
  - the discovered patterns and statistics **are further processed, filtered,** possibly resulting in aggregate user models that can be used as input to applications, such as recommendation engines, visualization tools, and Web analytics and report generation tools.

Web Usage Mining



Web Usage Mining

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- An important task in any data mining application is the creation of a suitable target data set to which data mining and statistical algorithms can be applied.
- This is particularly important in Web usage mining due to the characteristics of clickstream data and its relationship to other related data collected from multiple sources and across multiple channels.
- The process may involve **pre-processing** the original data, **integrating** data from multiple sources, and **transforming** the integrated data into a form suitable for input into specific data mining operations.

Web Usage Data Pre-Processing

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Data Fusion and Cleaning

- **Data fusion** refers to the **merging of log files** from several Web and application servers.
- **Data cleaning** is usually site-specific, and involves tasks such as, removing extraneous references to embedded objects that may not be important for the purpose of analysis, including references to style files, graphics, or sound files.

Web Usage Data Pre-Processing

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Pageview Identification

- Each **pageview** can be viewed as a collection of Web objects or resources representing a specific "user event".
- **Identification** of pageviews is heavily dependent on the **intra-page** structure of the site, as well as on the page contents and the underlying site domain knowledge.
- It may be desirable to consider pageviews at a **higher level** of aggregation.

Web Usage Data Pre-Processing

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User Identification

- The analysis of Web usage **does not require** knowledge about a user's **identity**.
- However, it is necessary to distinguish among **different users**.
- Since a user may visit a site more than once, the server logs **record multiple sessions** for each user.
- **IP addresses, alone**, are not generally sufficient for mapping log entries onto the set of unique visitors.
- **A combination** of IP addresses and other information, such as user agents and referrers.

Web Usage Data Pre-Processing

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Same IP, but different Agent

A combination of IP addresses and other information such as user agents and referrers.

Time	IP	URL	Ref	Agent
0:01	1.2.3.4	A	-	IE5;Win2k
0:09	1.2.3.4	B	A	IE5;Win2k
0:10	2.3.4.5	C	-	IE6;WinXP;SP1
0:12	2.3.4.5	B	C	IE6;WinXP;SP1
0:15	2.3.4.5	E	C	IE6;WinXP;SP1
0:19	1.2.3.4	C	A	IE5;Win2k
0:22	2.3.4.5	D	B	IE6;WinXP;SP1
0:22	1.2.3.4	A	-	IE6;WinXP;SP2
0:25	1.2.3.4	E	C	IE5;Win2k
0:25	1.2.3.4	C	A	IE6;WinXP;SP2
0:33	1.2.3.4	B	C	IE6;WinXP;SP2
0:58	1.2.3.4	D	B	IE6;WinXP;SP2
1:10	1.2.3.4	E	D	IE6;WinXP;SP2
1:15	1.2.3.4	A	-	IE5;Win2k
1:16	1.2.3.4	C	A	IE5;Win2k
1:17	1.2.3.4	F	C	IE6;WinXP;SP2
1:26	1.2.3.4	F	C	IE5;Win2k
1:30	1.2.3.4	B	A	IE5;Win2k
1:36	1.2.3.4	D	B	IE5;Win2k

User 1

0:01	1.2.3.4	A	-
0:09	1.2.3.4	B	A
0:19	1.2.3.4	C	A
0:25	1.2.3.4	E	C
1:15	1.2.3.4	A	-
1:26	1.2.3.4	F	C
1:30	1.2.3.4	B	A
1:36	1.2.3.4	D	B

User 2

0:10	2.3.4.5	C	-
0:12	2.3.4.5	B	C
0:15	2.3.4.5	E	C
0:22	2.3.4.5	D	B

User 3

0:22	1.2.3.4	A	-
0:25	1.2.3.4	C	A
0:33	1.2.3.4	B	C
0:58	1.2.3.4	D	B
1:10	1.2.3.4	E	D
1:17	1.2.3.4	F	C

Web Usage Data Pre-Processing

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Sessionization

- Sessionization is the process of segmenting the user activity record of each user into **sessions**, each representing **a single visit** to the site.
- The **goal** of a sessionization heuristic is to **reconstruct**, the actual sequence of actions performed by one user during one visit to the site, from the clickstream data.
- Sessionization heuristics fall into two basic categories:
  - **time-oriented:** apply either **global** or **local** time-out estimates to distinguish between consecutive sessions
  - **structure-oriented:** use either the **static site structure** or the **implicit linkage structure** captured in the referrer fields of the server logs

Web Usage Data Pre-Processing

User 1

Time	IP	URL	Ref
0:01	1.2.3.4	A	-
0:09	1.2.3.4	B	A
0:19	1.2.3.4	C	A
0:25	1.2.3.4	E	C
1:15	1.2.3.4	A	-
1:26	1.2.3.4	F	C
1:30	1.2.3.4	B	A
1:36	1.2.3.4	D	B

Session 1

0:01	1.2.3.4	A	-
0:09	1.2.3.4	B	A
0:19	1.2.3.4	C	A
0:25	1.2.3.4	E	C

Session 2

1:15	1.2.3.4	A	-
1:26	1.2.3.4	F	C
1:30	1.2.3.4	B	A
1:36	1.2.3.4	D	B

- ***h1:*** Total session duration may not exceed a threshold θ. Given t_0, the timestamp for the first request in a constructed session *S*, the request with a timestamp *t* is assigned to *S*, iff $t - t_0 \leq \theta$
- ***h2*:** Total time spent on a page may not exceed a threshold δ. Given t_1, the timestamp for request assigned to constructed session *S*, the next request with timestamp t_2 is assigned to *S*, iff $t_2 - t_1 \leq \delta$
- Example above: $\theta = 30$ mins
- If we apply **h2 with $\delta = 10$** mins, there are three sessions for User 1: **A→B → C → E**, **A**, and **F → B → D**

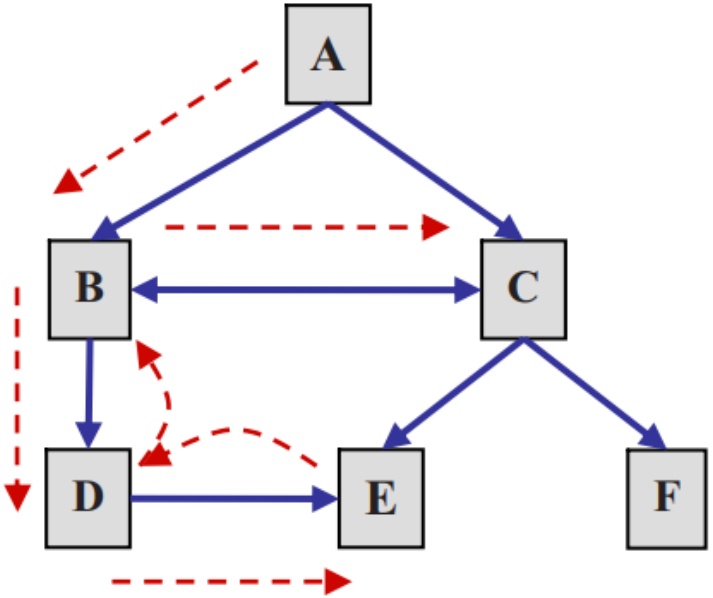
Web Usage Data Pre-Processing

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Path Completion

- Another potentially important pre-processing task which is usually performed after sessionization is **path completion**.
- Client- or proxy-side **caching** can often result in missing access references to those pages or objects that have been cached.
- **Missing references** due to caching can be heuristically inferred through **path completion,** which relies on the knowledge of site structure and the referrer information from server logs.

Web Usage Data Pre-Processing



User's actual navigation path:

A → B → D → E → D → B → C

What the server log shows:

URL	Referrer
A	--
B	A
D	B
E	D
C	B

Path Completion is needed, because of caching.

Web Usage Data Pre-Processing

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Data Integration

- In order to provide the most effective framework for pattern discovery, data from **a variety of other sources** must be integrated with the preprocessed clickstream data.
- For example, in e-commerce applications:
  - **user data**
  - **product attributes** and **categories** from operational databases.
  - **shopping cart changes**,
  - **order** and **shipping information**,
  - **impressions**,
  - **click-throughs**.

Web Usage Mining: Modelling

- Data Modelling for Web Usage Mining

Sessions / users	Pageviews A	B	C	D	E	F
user0	15	5	0	0	0	185
user1	0	0	32	4	0	0
user2	12	0	0	56	236	0
user3	9	47	0	0	0	134
user4	0	0	23	15	0	0
user5	17	0	0	157	69	0
user6	24	89	0	0	0	354
user7	0	0	78	27	0	0
user8	7	0	45	20	127	0
user9	0	38	57	0	0	15

user-pageview matrix Y

	A.html	B.html	C.html	D.html	E.html
user1	1	0	1	0	1
user2	1	1	0	0	1
user3	0	1	1	1	0
user4	1	0	1	1	1
user5	1	1	0	0	1
user6	1	0	1	1	1

	web	data	mining	business	intelligence	marketing	ecommerce	search	information	retrieval
user1	2	1	1	1	2	2	1	2	3	3
user2	1	1	1	2	3	3	1	1	2	2
user3	2	3	3	1	1	1	2	1	2	2
user4	3	2	2	1	2	2	1	2	4	4
user5	1	1	1	2	3	3	1	1	2	2
user6	3	2	2	1	2	2	1	2	4	4

content-enhanced transaction matrix X=Y*Z^T

	A.html	B.html	C.html	D.html	E.html
web	0	0	1	1	1
data	0	1	1	1	0
mining	0	1	1	1	0
business	1	1	0	0	0
intelligence	1	1	0	0	1
marketing	1	1	0	0	1
ecommerce	0	1	1	0	0
search	1	0	1	0	0
information	1	0	1	1	1
retrieval	1	0	1	1	1

term-pageview matrix Z

World-Wide-Web Mining

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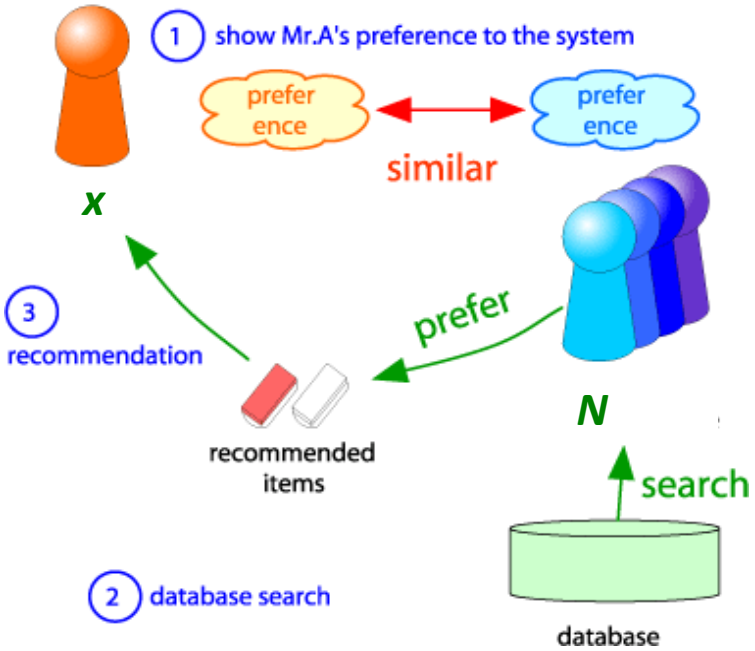
Recommender Systems and Collaborative Filtering

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- Recommender systems are widely used on the Web for **recommending products** and **services** to users. Most e-commerce Web sites have such systems.
- Two important functions:
  - help **users** deal with the information overload by giving them **personalized recommendations**.
  - help **businesses** make more profits.
- Example: **Movie recommendation** based on users' preferences and historical records.

Collaborative Filtering

- Consider user ***x***
- Find set ***N*** of other users whose ratings are "**similar**" to ***x***'s ratings
- Estimate ***x***'s ratings based on ratings of users in ***N***



Finding "Similar" Users

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

- Consider Users A and B with ratings above
- We need a similarity metric sim(A,B)
- Capture intuition that sim(A,B) > sim(A,C)

Option 1: Jaccard Similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

- **Sim(A,B) =** $\frac{|r_A \cap r_B|}{|r_A \cup r_B|}$
- **Sim(A,B) = 1/5; sim(A,C) = 2/4**
  - **Sim(A,B) < sim(A,C)**
  - **Problem:** Ignores the value of the rating

r_A = {4, 0, 0, 5, 1, 0, 0}
r_B = {5, 5, 4, 0, 0, 0, 0}
r_C = {0, 0, 0, 2, 4, 5, 0}

Option 2: Cosine Similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

r_A = {4, 0, 0, 5, 1, 0, 0}
r_B = {5, 5, 4, 0, 0, 0, 0}
r_C = {0, 0, 0, 2, 4, 5, 0}

- **Sim(A,B) = cos(A,B) = $\frac{dot(rA, rB)}{|r_A||r_B|}$**
- **Sim(A,B) = 0.38; sim(A,C) = 0.32**
  - **Sim(A,B) > sim(A,C), but not by much**

Option 3: Centered Cosine Similarity

- **Normalize ratings by subtracting row mean**

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

row_A = 10/3
row_B = 14/3
row_C = 11/3
row_D = 6/2 =3

- **Sim(A,B) = cos(A,B) = $\frac{dot(rA,rB)}{|A||B|}$**
- **Sim(A,B) = 0.09; sim(A,C) = -0.56**
  - **Sim(A,B) > sim(A,C)**
- **Missing ratings treated as "average"**
- **Also known as Pearson Correlation**

r_A = {2/3, 0, 0, 5/3, -7/3, 0, 0}
r_B = {1/3, 1/3, -2/3, 0, 0, 0, 0}
r_C = {0, 0, 0, -5/3, 1/3, 4/3, 0}

Rating Predictions: User-User

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From similarity metric to recommendations:

- Let $\boldsymbol{r_x}$ be the vector of user $\boldsymbol{x}$'s ratings
- Let $\boldsymbol{N}$ be the set of $\boldsymbol{k}$ users most similar to $\boldsymbol{x}$ who have rated item $\boldsymbol{i}$
- **Prediction for item *i* of *user x*:**
  - Option 1: $r_{xi} = \frac{1}{k} \sum_{y \in N} r_{yi}$ (k-NN)
  - Option 2: $r_{xi} = \frac{\sum_{y \in N} s_{xy} \cdot r_{yi}}{\sum_{y \in N} s_{xy}}$

Shorthand:

$s_{xy} = sim(x, y)$

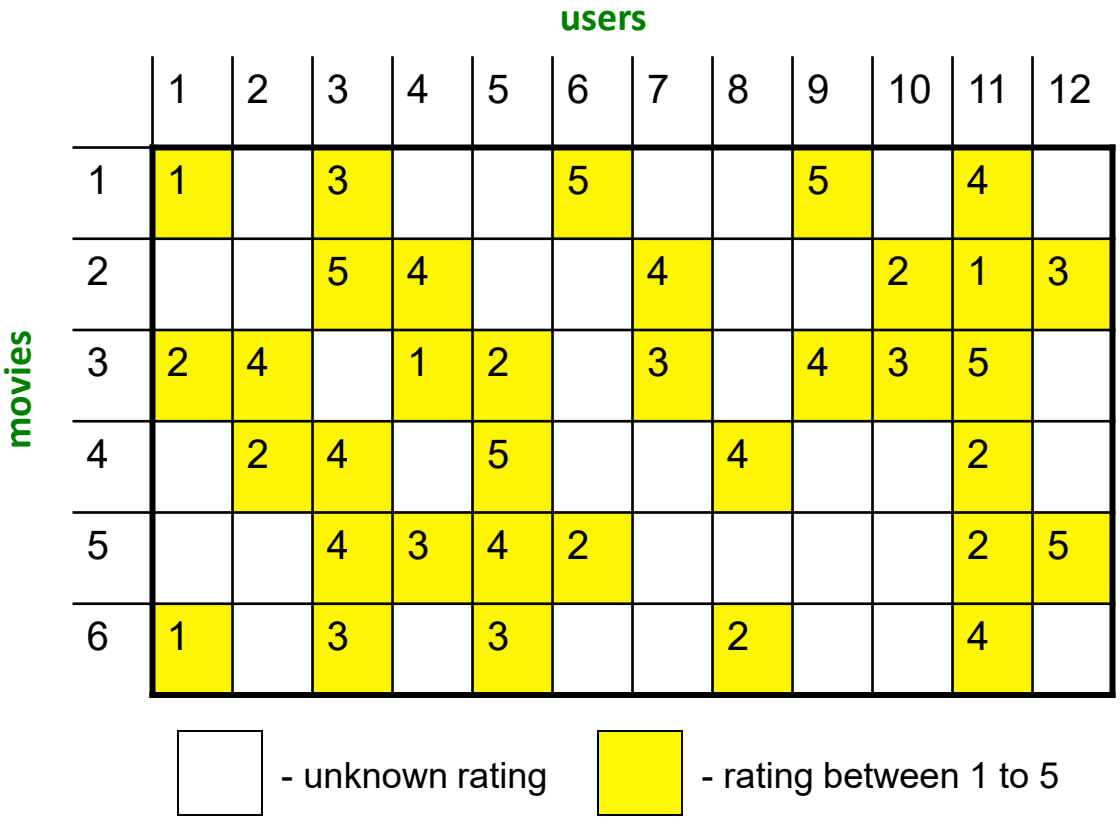
Item-Item Collaborative Filtering

- **So far: User-user collaborative filtering**
- **Another view: Item-item**
  - For item ***i***, find other similar items (items in neighborhood)
  - Estimate rating for item ***i*** based on ratings for similar items
  - Use the same similarity metrics and prediction functions as in user-user model

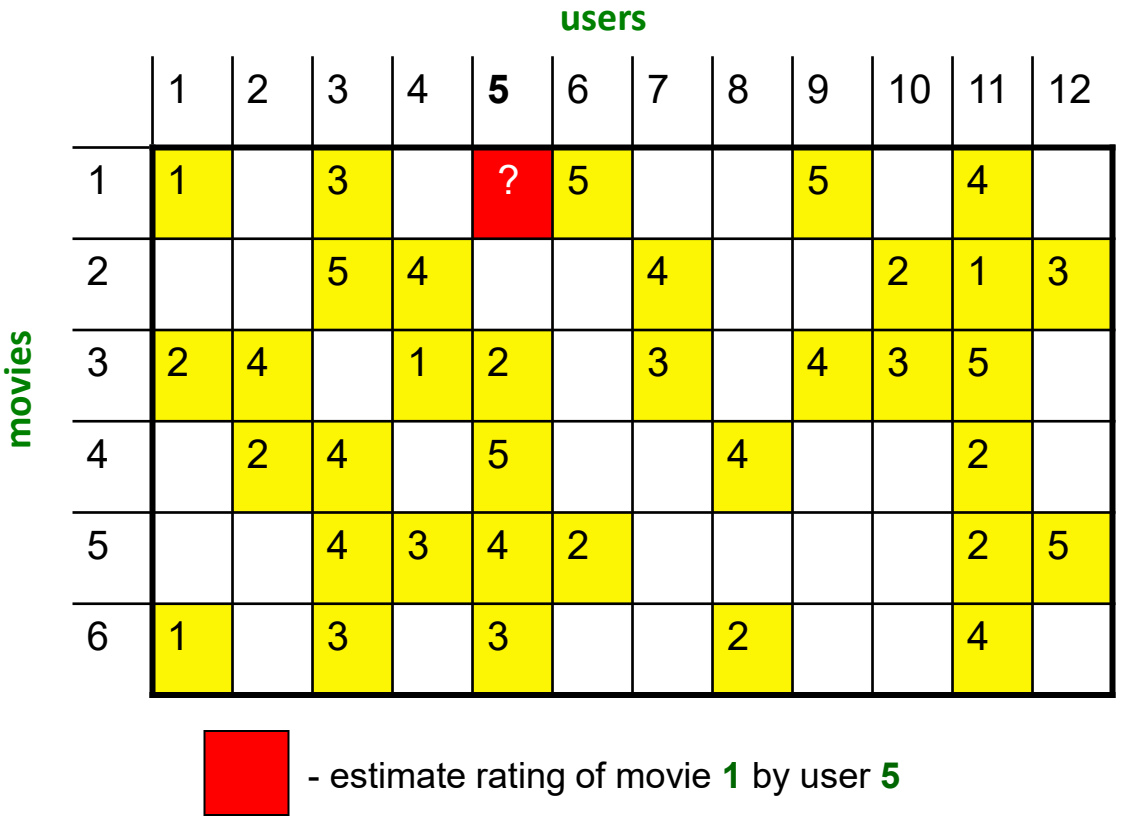
$$r_{xi} = \frac{\sum_{j \in N(i;x)} s_{ij} \cdot r_{xj}}{\sum_{j \in N(i;x)} s_{ij}}$$

s_{ij}… similarity of items ***i*** and ***j***
r_{xj}…rating of user ***u*** on item ***j***
$N(i;x)$… set of items rated by ***x*** similar to ***i***

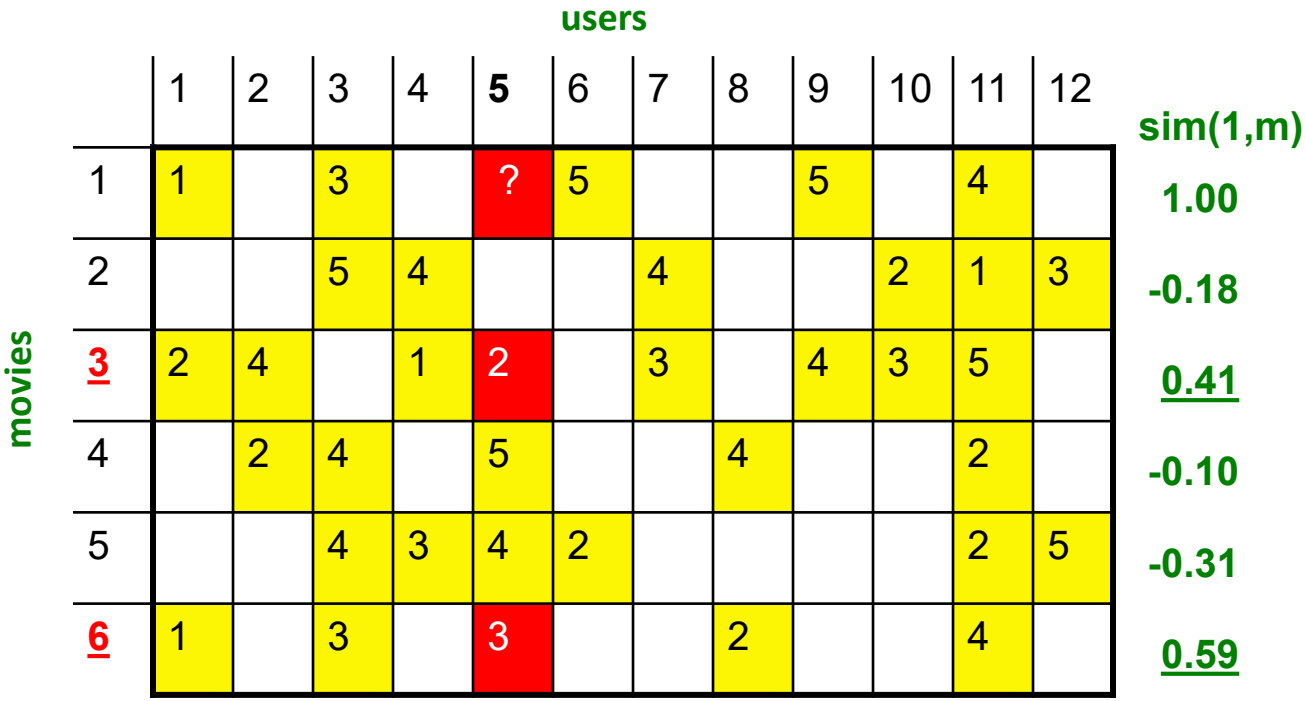
Item-Item CF (|N|=2)



Item-Item CF (|N|=2)



Item-Item CF (|N|=2)



Neighbor selection:
Identify movies similar to movie **1**, **rated by user 5**

Here we use Pearson correlation as similarity:

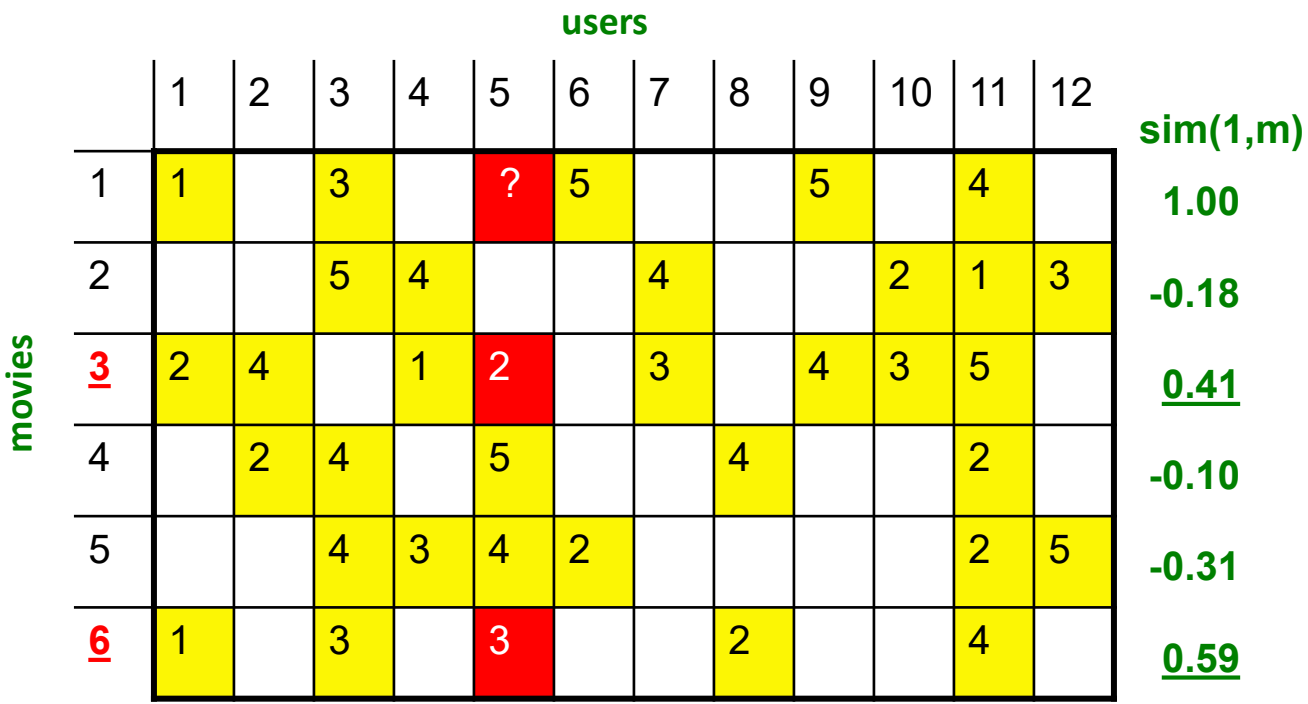
1) Subtract mean rating m_i from each movie i

$m_1 = (1+3+5+5+4)/5 = 3.6$

row 1: *[-2.6, 0, -0.6, 0, 0, 1.4, 0, 0, 1.4, 0, 0.4, 0]*

2) Compute cosine similarities between rows

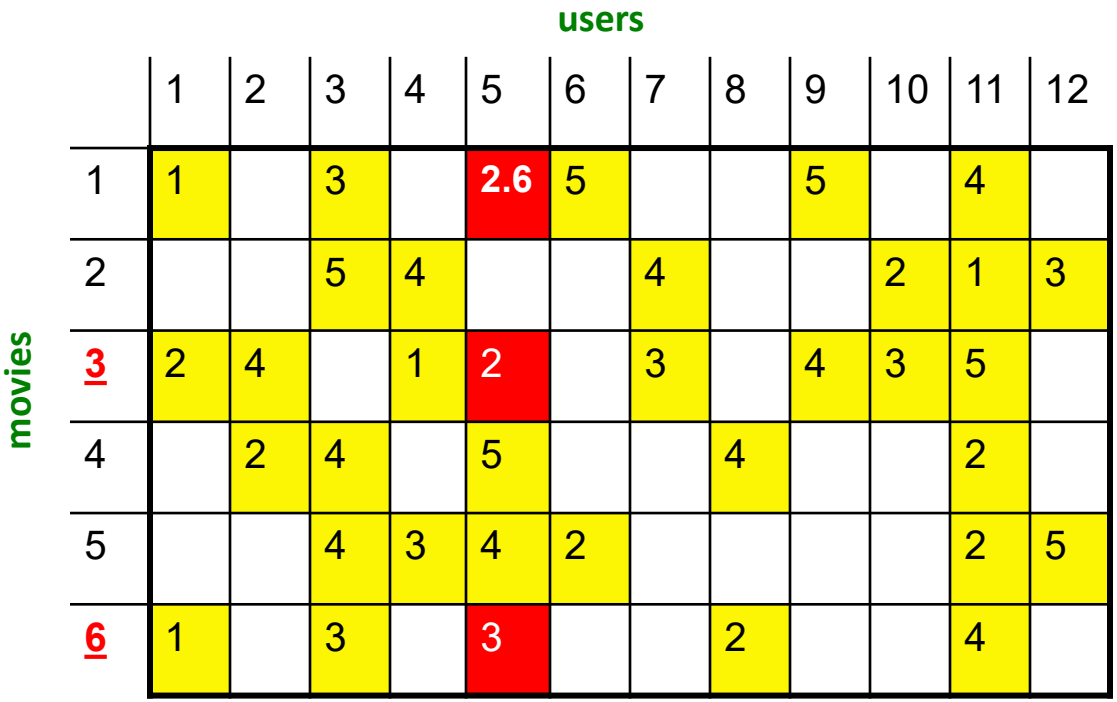
Item-Item CF (|N|=2)



Compute similarity weights:

$s_{1,3}$=0.41, $s_{1,6}$=0.59

Item-Item CF (|N|=2)



Predict by taking weighted average:

$r_{1.5}$ = (0.41*2 + 0.59*3) / (0.41+0.59) = 2.6

$$r_{ix} = \frac{\sum_{j \in N(i;x)} s_{ij} \cdot r_{jx}}{\sum s_{ij}}$$

Item-Item vs. User-User

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- In theory, **user-user** and **item-item** are dual approaches
- In practice, item-item **outperforms** user-user in many use cases
- **Why**? Items are simpler, users have multiple **tastes**.
  - Items belong to a small set of "genres", but users have varied tastes
  - Item Similarity is more meaningful than User Similarity

References

Yehuda Koren, Robert Bell, Chris Volinsky. Matrix Factorization Techniques for Recommender Systems. IEEE Computer 42 (8), pp. 30-37, 2009.

Pros/Cons of Collaborative Filtering

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- **+ Works for any kind of item**
  - No feature selection needed
- **- Cold Start:**
  - Need enough users in the system to find a match
- **- Sparsity:**
  - The user/ratings matrix is sparse
  - Hard to find users that have rated the same items
- **- First rater:**
  - Cannot recommend an item that has not been previously rated
  - New items, Esoteric items
- **- Popularity bias:**
  - Cannot recommend items to someone with unique taste
  - Tends to recommend popular items

Summary

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Have a question?

